

# **Mandatory HC-POD Final Integration**

## **Track Collaborated With**

### **Data Science**

#### **1. Dependency**

The Data Analytics track depended on the Data Science track to assess whether patterns identified during descriptive and exploratory analysis could support predictive modelling and to provide model-based evidence to compare with validated analytical findings.

The Data Science track relied on Data Analytics for validated variables, KPI definitions, identified patterns, and interpretation of important appointment characteristics to inform model development and interpretation.

#### **2. Output Received**

Data Analytics received the relevant Data Science modelling output, including the model findings and interpretation of the variables/features considered important for predicting appointment no-shows.

The received output was reviewed against the validated Data Analytics findings relating to:

- Previous no-show behaviour
- Booking lead time
- Distance to clinic
- Reminder channel
- Appointment type
- Appointment outcome/no-show status

The purpose of the comparison was to determine whether the predictive analysis supported, differed from, or added context to the descriptive findings.

#### **3. Output Provided**

Data Analytics provided the Data Science intern with validated analytical findings and supporting definitions, including:

- **5,000 total appointments**
- **2,423 no-shows**
- **2,314 attended appointments**
- **263 cancellations**
- **48.46% no-show rate**
- **46.28% attendance rate**
- **5.26% cancellation rate**
- Previous no-show comparison
- Booking lead-time analysis
- Distance-to-clinic analysis

- Reminder-channel analysis
- Appointment-type analysis
- Relevant limitations concerning observational analysis and causal interpretation

Data Analytics also highlighted that the dashboard KPI required correction: **5.26% represents the cancellation rate, not the no-show rate. The validated no-show rate is 48.46%.**

#### 4. Final Integration Activity

The final integration activity involved comparing the validated Data Analytics findings with the Data Science model output and using the results to strengthen the final HealthConnect decision-support solution.

The process was:

**Data Analytics findings → Data Science modelling → model output → cross-check against analytical findings → refinement of final insights → final integrated HealthConnect presentation**

The Data Analytics findings were therefore not treated as a standalone dashboard exercise. They were used as an analytical reference point for interpreting the predictive modelling results.

#### 5. What Changed

The collaboration changed the final solution in three main ways.

**First**, the final analysis was positioned as a combination of descriptive analytics and predictive analysis rather than relying only on historical patterns.

**Second**, variables identified through Data Analytics as potentially relevant to no-show behaviour—particularly previous no-show history, booking lead time and distance—could be considered alongside the Data Science model findings when explaining potential risk factors.

**Third**, the final dashboard and reporting were refined to ensure that KPI definitions and analytical interpretations were consistent before being incorporated into the final HealthConnect presentation.

The no-show KPI was specifically corrected during the validation process:

**No-show rate = 2,423 / 5,000 = 48.46%**

while:

**Cancellation rate = 263 / 5,000 = 5.26%**

#### 6. How the Change Improved the Solution

The integration improved the HealthConnect solution by connecting **historical/descriptive evidence with predictive analysis.**

Instead of presenting isolated findings, the final solution provides a clearer analytical pathway:

**What happened?**

Data Analytics identified patterns in appointment outcomes.

**Which factors may be useful for prediction?**

Data Science assessed relevant variables through predictive modelling.

**How should the findings be interpreted?**

The two analytical perspectives were compared before being incorporated into the final decision-support output.

This makes the final solution more evidence-based and gives HealthConnect a stronger basis for investigating appointment no-shows and potential intervention strategies.

## **7. Evidence**

Evidence of the integration includes:

- Final Data Analytics dashboard
- Validated KPI calculations
- Data Analytics findings shared with Data Science
- Data Science modelling output received
- Comparison of Data Analytics findings with Data Science results
- Final refined analytical findings
- Updated dashboard/report containing the corrected KPI definitions
- HC-POD collaboration/integration record
- Final presentation slides showing the relationship between descriptive and predictive analytics

Where available, screenshots of the exchanged outputs and the final integrated dashboard should be attached to the HC-POD evidence folder.

## **8. Contribution to the Final Presentation**

The Data Analytics contribution to the final HealthConnect presentation will cover:

1. Overall appointment outcomes and validated KPIs.
2. Key no-show patterns identified through analysis.
3. Important segmentation findings, including previous no-show behaviour, booking lead time and distance.
4. Relevant limitations and interpretation of the analytical findings.
5. The connection between Data Analytics findings and Data Science predictive modelling.
6. The corrected final no-show and cancellation KPIs.
7. Business recommendations arising from the integrated analysis.

The presentation will therefore demonstrate how Data Analytics and Data Science contributed to one coherent HealthConnect solution rather than presenting two independent track outputs.

## **Final HC-POD Integration Statement**

The Data Analytics track collaborated with the Data Science intern through an actual exchange of analytical findings, modelling output and interpretation. Data Analytics supplied validated appointment-outcome metrics and identified potentially relevant no-show patterns, while Data Science provided predictive modelling evidence that was reviewed against those findings. The final integration combined descriptive and predictive perspectives, corrected the KPI interpretation, strengthened the explanation of potential no-show risk factors, and provided a more coherent evidence base for the final HealthConnect decision-support solution.